

YASH KASHYAP

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EDUCATION

Vellore Institute of Technology

B.Tech Computer Science Engineering

Kendriya Vidyalaya

Class 12 – 93.6% — Class 10 – 93.8%

2023 – 2027

CGPA: 9.56/10

Senior Secondary

EXPERIENCE

Web Development Intern

Summer 2025

NIT Rourkela – Full Stack + AI/ML

[Certificate]

- Engineered a full-stack team management platform (Next.js, TypeScript, Supabase) for **200+ participants** with Google OAuth restricted to `@vitstudent.ac.in`, **3-level RBAC**, and slot-capacity rule.
- Developed a traffic density prediction platform (React, Django REST, PostgreSQL) using a Random Forest model with **81.5% accuracy** across 62 locations in Delhi.

PROJECTS

Sentinel – AI Model Auth via Behavioral Fingerprinting | *FastAPI, PostgreSQL, Redis, Celery* 🔒

- Designed a runtime authentication system detecting ML model swaps via **7 REST/WebSocket endpoints**; load-tested at **50 concurrent users** with $\geq 85\%$ **test coverage** across core services.
- Architected a **3-service pipeline** (Probe Engine, Fingerprint Vault, Runtime Verifier) with scheduled probing and live WebSocket score streaming.

Hybrid Leak Detection System | *Python, Scikit-learn, Django, Streamlit* 🔒

- Built a **6-model ensemble** (4 Autoencoders, OC-SVM, Mahalanobis) for diesel engine leak detection on **40K samples** across 12 features, with a real-time WebSocket pipeline and Streamlit dashboard.
- Achieved zone-level leak isolation with **macro F1: 0.884** via Kalman filtering and physics-based discriminators.

DevSecOps – Repository Security & Compliance Scanner | *React, Django REST, Celery, Supabase* 🔒

- Implemented a platform with **10 REST endpoints** scanning GitHub repos across **4 security dimensions**; async Celery + Redis pipeline achieved **sub-2s response times** with a 120s scan timeout.
- Integrated pip-audit, bandit, and radon scanners with JWT auth via Supabase and a 3D radar visualization.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript

Web & Backend: React.js, Node.js, Express.js, Django REST Framework, FastAPI, HTML5, CSS3

AI/ML: Scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib, Supervised & Unsupervised Learning, NLP

Databases & Tools: PostgreSQL, MySQL, MongoDB, Supabase, Git, Docker, Vercel

LEADERSHIP

Technical Head

Aug 2025 – Present

Soft Computing Research Society (SCRS), VIT Vellore

- Spearheaded the **Escape.exe** event during Gravitas, integrating a custom-built full-stack team management platform to streamline registrations and seamlessly manage participant workflows.
- Coordinated **From URL to Response**, an online technical session breaking down the complete lifecycle of a web request from browser to server to database.

CERTIFICATIONS

- AWS Certified Cloud Practitioner** (Amazon Web Services) [Certificate]
- Oracle Gen AI Professional** (Oracle) [Certificate]
- Generative AI with Watsonx** (IBM) [Certificate]
- Machine Learning** (Stanford Online) [Certificate]
- Full Stack Web Dev** (Udemy) [Certificate]